Standard Operating Procedure for Resolving Critical BGP Route Churn, OSPF Adjacency Loss, and ISIS LSP Flooding on Arista Access Node IND-ACC-76 at Indiranagar Exchange

Node: IND-ACC-76 | **Vendor:** Arista | **Severity:** Critical | **Alarm Code:** ALARM_1665

Description

BGP route churn, OSPF adjacency loss, and ISIS LSP flooding on Arista access nodes can occur due to various reasons, including misconfiguration, hardware defects, or transient network events. In the context of Node ID IND-ACC-76 at the Indiranagar Exchange, this failure domain is a critical concern that requires immediate attention to prevent network instability and potential data loss. The Arista EOS (Extensible Operating System) is a complex software system that manages the control and data planes of the access node. The BGP routing protocol is used for exterior routing, while OSPF and ISIS are used for interior routing. The access node's configuration and hardware must be carefully examined to identify the root cause of the failure domain. This SOP outlines the steps to diagnose and resolve the issue, ensuring that the access node returns to a stable state and normal network operations are restored.

Pre-requisite Checks

- Ensure that all personnel involved in the troubleshooting process are properly trained and familiar with Arista EOS and the access node's configuration.

- Verify that the access node is powered on and all fans are operational.

- Check the redundant link status and ensure that it is functioning correctly.

- Dispatch remote hands to the Indiranagar Exchange to assist with physical checks and equipment replacement if necessary.

Potential Root Causes

Cause	Probability
BGP route churn due to misconfigured neighbor	High
OSPf adjacency loss due to network topology changes	Medium
ISIS LSP flooding due to hardware defects	Low

Diagnostic Commands & Expected Outputs

```
Command: show bgp summary, Expected Output: BGP table version is 123, Description: This command displays a summary of the BGP table, including the table version, number of routes, and neighbor status. It
```

```
helps identify any issues with the BGP neighbor configuration.

Command: show ospf neighbor, Expected Output: OSPF neighbor table is empty, Description: This command displays the OSPF neighbor table, which can help identify any issues with OSPF adjacency formation or neighbor relationships.

Command: show isis database, Expected Output: ISIS LSP database is empty, Description: This command displays the ISIS LSP database, which can help identify any issues with ISIS LSP flooding or database corruption.
```

Resolution Steps

Step 1: Step: Verify BGP neighbor configuration, Command: show bgp neighbors, Description: Check the BGP neighbor configuration to ensure that it is correct and consistent with the network design. If the configuration is incorrect, correct it and verify that the issue is resolved.

Step 2: Step: Reset OSPF adjacency, Command: clear ospf neighbor, Description: Reset the OSPF adjacency by clearing the OSPF neighbor table and re-establishing the adjacency. This can help resolve issues with OSPF adjacency formation or neighbor relationships.

Step 3: Step: Flush ISIS LSP database, Command: clear isis database, Description: Flush the ISIS LSP database to resolve issues with ISIS LSP flooding or database corruption.

Post-Check Verification

```
Command: show bgp summary, Expected Output: BGP table version is 123, Description: Verify that the BGP table version is correct and that the number of routes is accurate.

Command: show ospf neighbor, Expected Output: OSPF neighbor table is populated, Description: Verify that the OSPF neighbor table is populated and that the neighbor relationships are correct.

Command: show isis database, Expected Output: ISIS LSP database is populated, Description: Verify that the ISIS LSP database is populated and that there are no issues with ISIS LSP flooding or database corruption.
```

Roll-back Procedures

- Step: Restore BGP neighbor configuration, Command: load merge /tmp/bgp_neighbors.conf, Description: Restore the BGP neighbor configuration from the backup configuration file.

- Step: Reset OSPF adjacency, Command: clear ospf neighbor, Description: Reset the OSPF adjacency to restore the previous state.

- Step: Flush ISIS LSP database, Command: clear isis database, Description: Flush the ISIS LSP database to restore the previous state.